

Mocaverse code scanning

Overall assessment

Development quality (Low)

- most of the code is forked from greenfield and evmos
 - there are too much outdated evm security/features compared with current cosmos/evm
- currently there are 2-3 devs working on this repo, but the way they work looks not efficient and reliable.
 - peer code review quality is bad

Current status

- (test-purpose) mainnet ([explorer](#))
 - v1.2.1
 - [genesis](#)
 - stablecoins, bridge (stargate, wormhole) deployed
- testnet ([explorer](#))
 - v1.2.0-rc2
 - [genesis](#)
 - AirKit(DID) is deployed
 - bucket NFT barely exist
- past security audit on v1.1.0
- ▼ modules of both mainnet and testnet
 - evm functionality - forked from evmos v12 (latest version is v20 and this version was released in 2024 and isn't maintained anymore)
 - evm
 - erc20
 - feemarket

- ibc
- storage layer functionality - forked from greenfield (bnb chain)
 - payment
 - permission
 - sp
 - storage
 - virtualgroup
 - challenge
 - *bridge, epochs, and oracle (these are deleted in the main branch, they're relevant to bnb chain)*

▼ chain metrics

metric	mainnet	testnet
total blocks	11.2M	10.4M
total txs	175K	390K
addresses	2,914	44,274
verified contracts	~43	~50
ERC-721 tokens	0	6 (incl. Bucket NFT, MocaNFT)
most active contract	PriceFeed (159K txs)	-

Mocachain features

- DID
 - AIRKit = Account, Identity, Reputation Kit (TypeScript SDK: `@mocanetwork/airkit`)
 - contracts deployed on **testnet only**: IssuerStakingController, PaymentsController, VotingController, EscrowedMoca
- storage layer
 - S3-like bucket provider with onchain metadata for each file and bucket
 - buckets and objects represented as NFTs on-chain
 - storage providers (SPs) manage off-chain data, coordinated via virtual groups

- streaming payment model (per-second billing)
- challenge system for data integrity verification (BLS-aggregated attestation)
- evm

EVM status

critical CVEs that requires patch

CVE	Severity	Issue	Impact
CVE-2024-32644	CRITICAL	Precompile stateDB.Commit() during execution allows partial state persistence	Infinite token minting (PoC: 1 ETH → 1M ETH in 20 iterations)
CVE-2024-37153	CRITICAL	ICS-20 transfer from smart contract doesn't deduct contract balance	Infinite money glitch via IBC transfers
GHSA-3fp5-2xwh-fxm6	CRITICAL	Non-atomic A→B→A state transitions bypass persistence	Same class as infinite mint
GHSA-mjfq-3qr2-6g84	HIGH	Low gas limit causes partial precompile execution without revert	Fund theft, validator halt

missing security features

- no single-EVM-tx-per-Cosmos-tx enforcement - allows batching attacks
- no atomic precompile execution (RevertMultiStore()) - fundamental safety gap
- no IBC rate-limit middleware - no protection against rapid fund drainage
- non-deterministic state mutations in EVM pre-blocker - consensus fork risk

missing evm features

- eip7702
- custom evm app side mempool
- ...

